HAO "TROY" YU

Address: Rego Park, NY **Portfolio Website:** troyyu527.github.io/ReactWebsite
Mobile: 347.634.1025 **Email:** troy0527@gmail.com

Software Engineer | BIM & CAD Automation Developer

AREAS OF EXPERTISE

Engineering Software Development ~ BIM & CAD Automation And Delivery ~ AEC API Development
Desktop / Web Application Development ~ AI-Assisted Engineering Workflows ~ Cloud & Data Integration

TECHNICAL ACUMEN

C# | .NET Framework | WPF | MVVM | Python | JavaScript | React | Node.js | D3.js | Three.js
Revit API | AutoCAD API | Civil3D API | Tekla Open API
Azure DevOps | Git | SQL | MySQL | ACC | BIM 360
Prompt Engineering | Large Language Models (LLMs) | Retrieval-Augmented Generation (RAG)
Agentic Workflows | AI-Assisted Development | Local AI Deployment | Ollama | MCP

RELEVANT EXPERIENCE

BlueScope Buildings North America, Inc. • Kansas City, MO, USA
Software Engineer (Contract via Innova Solutions) 01/2026 – 06/2026

- **Developed enterprise-grade engineering software applications and Tekla Structures plugins using C#, .NET Framework, WPF, MVVM, and Tekla Open API.**
- **Designed and implemented engineering automation features for steel building workflows, reducing manual design and detailing effort through API-driven design automation solutions.**
- **Developed reusable ViewModels, UI components, and application frameworks following WPF MVVM architecture and enterprise software design patterns.**
- **Contributed to the migration of legacy engineering applications into BlueScope's next-generation plugin platform, improving maintainability, scalability, and long-term product support.**
- **Leveraged AI-assisted development tools and Large Language Models (LLMs) to support code generation, code review, technical research, debugging, documentation, and software migration activities, improving development productivity and knowledge transfer.**
- **Participated in software architecture discussions, solution design reviews, sprint planning, and technical decision-making activities within an Agile Scrum environment.**

AECOM • New York, NY, USA
BIM Supervisor – Digital Solutions 01/2024 – 12/2025

- **Spearheaded the development and implementation of BIM standards, digital delivery workflows, and cloud-based collaboration strategies across multidisciplinary project teams.**
- **Evaluated and integrated emerging BIM technologies, automation tools, APIs, and cloud-based solutions to improve engineering productivity, data accessibility, and project delivery efficiency.**
- **Key Automation & Digital Solutions Contributions:**
  - **Developed custom Revit API solutions and reusable C# libraries to automate data extraction, model analysis, and integration between CAD, BIM, and external systems.**
  - **Designed workflow automation solutions using Dynamo, SQL, and Excel, reducing manual processing effort and improving consistency of engineering deliverables.**
  - **Managed Autodesk Construction Cloud (ACC) deployments, streamlining model coordination, cloud collaboration, and data exchange workflows.**
  - **Researched and piloted emerging AEC technologies and automation initiatives to support digital transformation and future technology adoption.**
  - **Performed model validation, quality assurance, and BIM data management activities to ensure compliance with project standards, client requirements, and digital delivery objectives.**

Urbantech Consulting Engineering PC • New York, NY, USA
BIM Engineer 10/2014 – 12/2022

- **Led the company's transition from traditional CAD-based workflows to BIM-based project delivery, establishing BIM standards, modeling practices, coordination workflows, and digital information management processes across structural engineering projects.**
- **Utilized Autodesk Revit, AutoCAD, Civil3D, and Navisworks to support BIM coordination, multidisciplinary collaboration, engineering design, drafting, and project delivery activities.**
- **Key Automation & Software Development Contributions:**
  - **Developed custom Revit API, AutoCAD API, and Civil3D API solutions using C#, leveraging custom User Attributes and metadata structures to automate data extraction, model management, and interoperability between CAD and BIM platforms.**
  - **Created reusable software libraries, plugins, and desktop applications that streamlined engineering workflows, reduced repetitive tasks, and improved operational efficiency.**
  - **Designed automation tools for engineering and detailing processes, including rebar quantity takeoff, bill-of-material generation, Autodesk license management, and project data management solutions.**
  - **Established automation-driven workflows that improved engineering coordination, reduced manual data processing, and enhanced digital project delivery.**
  - **Implemented BIM workflows utilizing point cloud technologies, LiDAR scanning data, and model integration processes to support coordination, analysis, and digital delivery initiatives.**

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- **Researched, evaluated, and piloted emerging AEC technologies, automation solutions, and digital workflows to modernize engineering processes and improve productivity.**

SELECTED TECHNICAL PROJECTS

UNO Multiplayer Web Game (http://troyyu527.github.io/uno-web-game)
Next.js, Node.js, Express.js

- **Developed a full-stack multiplayer web application featuring user authentication, persistent game states, and RESTful API services.**
- **Implemented a heuristic-based AI player utilizing weighted decision-making and priority queue algorithms.**
- **Designed front-end and back-end architectures using Next.js, Express.js, JWT, Passport.js, and Bcrypt.**

Visualization of Data Structures & Sorting Algorithms (http://troyyu527.github.io/visual-algo)
D3.js, JavaScript

- **Developed interactive visualizations of trees, heaps, graphs, and sorting algorithms using D3.js and SVG rendering.**
- **Created reusable visualization modules and asynchronous event-driven interactions to improve user engagement and maintainability.**

Hide & Seek Pathfinding Visualizer (http://troyyu527.github.io/hidenseek)
React, JavaScript

- **Developed an interactive pathfinding visualization platform demonstrating Dijkstra's shortest-path algorithm and procedural maze generation.**
- **Designed reusable React component architecture and state management patterns supporting dynamic route analysis and visualization.**

ADDITIONAL RESEARCH EXPERIENCE

MSV (Mechanic Sound Vibration) Lab, National Taiwan Ocean University • Keelung, Taiwan
Research Assistant – Computational Analysis & Modeling 2009–2010

- **Developed engineering models using MATLAB for applied mathematics, with Finite Element Analysis and elements result computation in engineering mechanics problems.**
- **Worked on practical problems by formulating numerical equations and studying mathematical models.**

EDUCATION

Master of Science in Civil Engineering and Engineering Mechanics • 2014
Columbia University • New York, NY USA